

Second Terminal Examination - 2079

Class : XII

Subject: Physics (1021)

Full Marks: 75

Time: 3 hrs.

SET 'A'

Pass Marks: 30

Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

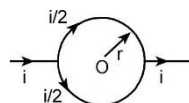
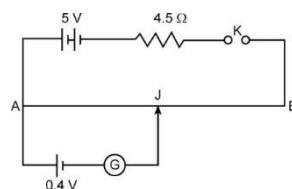
Attempt all the questions.

Group A

Rewrite the best alternative of each question in your answer sheet. [11 × 1 = 11]

- If a person standing on a rotating disc stretches out his hands, the angular speed will
 - increase
 - decrease
 - remains constant
 - none
- In SHM, the particle velocity is maximum at
 - mean position
 - extreme position
 - at a distance $\frac{A}{2}$
 - at a distance $\frac{A}{4}$
- A constant torque of 1000Nm turns a wheel of moment of inertia 200kgm² about its center. Its angular velocity after 3 second is
 - 1 rad/s
 - 5 rad/s
 - 10 rad/s
 - 15 rad/s
- Which relation is correct for adiabatic process?
 - $\Delta Q = \Delta W$
 - $\Delta Q = \Delta U$
 - $\Delta W = -\Delta U$
 - none of these

5. The efficiency of the Carnot engine operating between the steam point and ice point is
 a) 100% b) 73.2% c) 50% d) 26.8%
6. A progressive wave has the equation $y = 0.02\sin(30t - 4x)$, where x is in meters and t in seconds. The phase velocity is
 a) 7.5 m/s b) 6.5 m/s c) 5.5 m/s d) 4.5 m/s
7. The velocity of sound in air is double the velocity in air at 0°C at the temperature
 a) 278°C b) 819°C c) 1092°C d) 800°C
8. In figure, the potentiometer wire AB has a resistance of 5Ω and length 10 m. The balancing length AJ for the emf of 0.4 V is
 a) 4 m b) 1.5 m
 c) 0.8 m d) 0.4 m
9. The power of a bulb is 100 watts at 220 V. When the voltage is 110 V, the power of the bulb is:
 a) 150 W b) 50 W c) 120 W d) 25 W
10. A straight conductor carrying a direct current 'i' amp is split into a circular loop as shown in Fig. Then the magnetic induction at the centre of the circular loop of radius r metre is :
 a) 0 b) ∞ c) $\frac{\mu_0 i}{2\pi r}$ d) $\frac{\mu_0 i}{2 r}$
11. An electron of charge 'e' in rest in an electric field between two plates separated by a distance 'd' and with Volt difference 'V'. The force experienced by an electron is
 a) $\frac{eV}{d}$ b) $\frac{ed}{V}$ c) $\frac{d}{eV}$ d) $\frac{V}{d}$

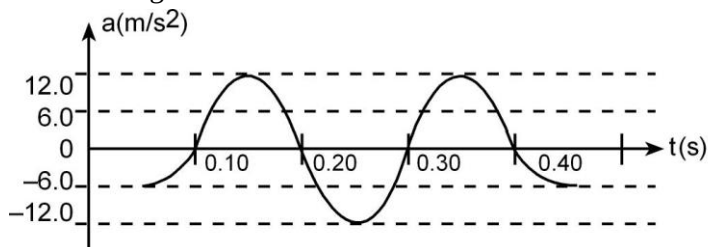


Group - B

Give short answer to the following questions.

[8 × 5 = 40]

12. a) Define moment of inertia. On what factors does it depend? [2]
b) Can the object rotate if there is no torque acting on it? Defend your answer. [1]
c) A ballet dancer spins at 1 revolution per second with her arms stretched. With her arms folded, her moment of inertia about the axis of rotation decreases by 40%. Calculate the ratio of her initial kinetic energy of rotation to the final kinetic energy of rotation. [2]
13. a) On a frictionless horizontal air track, a glider oscillates at the end of an ideal spring of force constant 2.50 N/cm. The graph shows the acceleration of glider as function of time



Calculate:

- i. mass of glider. [2]
ii. maximum displacement of glider from mean position. [1]
- b) If a pendulum clock is taken to the top of mountain, does it gain or lose time? Explain. [1]
- c) Can a pendulum clock be used in an artificial satellite? Explain. [1]

OR

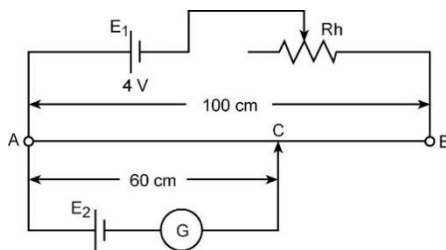
- a) What is simple harmonic oscillator? Obtain an expression for its total energy and time period. [4]
- b) A particle of mass 0.3kg vibrates with a period of 2 seconds. If the amplitude is 0.5m, what is its maximum kinetic energy? [1]

14. a) Can the Carnot's engine be realized in practice? On what factors does the efficiency of Carnot engine depend? [2]
- b) A Carnot engine has 40% efficiency with a sink at 10°C . By how many degrees should the temperature of the source be increased in order to raise the efficiency to 65%? [3]

OR

- a) During adiabatic changes, the volume of a gas is found to depend inversely on the squares of its absolute temperature.
- Find how its pressure will depend on the absolute temperature [1]
 - Calculate the ratio of specific heat capacity of gas at constant pressure to constant volume. [2]
- b) Show the slope of an adiabatic curve at any point is γ times the slope of an isothermal curve at the corresponding points. [2]
15. a) Distinguish between progressive wave and stationary wave. [2]
- b) Show that both odd and even harmonics are present in an open organ pipe. [3]
16. a) Why do we prefer a potentiometer to measure emf of a cell rather than a voltmeter? [2]

- b) A battery E_1 of 4 V and a variable resistance R_h are connected in series with the wire AB of the potentiometer. The length of the wire of the potentiometer is 1 meter. When

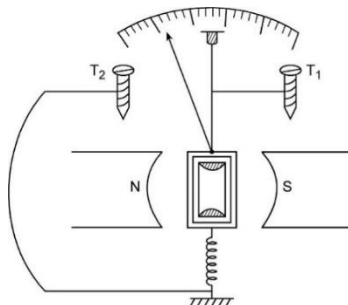


a cell E_2 of emf 1.5 volt is connected between points A and C, no current flows through E_2 . Length of AC = 60 cm.

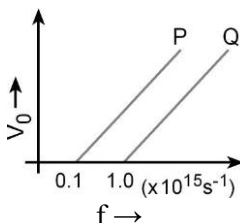
- Find the potential difference between the ends A and B of the potentiometer. [2]
- Would the method work, if the battery E_1 is replaced by a cell of emf of 1V? [1]

17. The schematic diagram of a moving coil galvanometer is shown in figure [3]

- a)
 - i. On what principle does it depend?
 - ii. Why do we make radial magnetic field?
 - iii. How can the current sensitivity of moving coil galvanometer be increased?
- b) A coil consisting of 100 circular loops with radius 60cm carries a current of 5A. Find the magnetic field at a point along the axis of the coil, 80cm from the center. ($\mu_0 = 4\pi \times 10^{-7} \text{ T m/A}$) [2]



18. a) Do all the electrons emitted in photoelectric effect have the same kinetic energy? Explain. [1]
- b) The work function of silver is 4.73eV. What does it mean? [1]
- c) Figure shows the variation of stopping potential (V_0) with the frequency (f) of the incident radiation for two photo sensitive metals P and Q.



- i. Explain which metal has smaller threshold wavelength. [1]
- ii. Explain, giving reason, which metals emits photoelectrons having smaller kinetic energy, for the same wavelength of incident radiation. [1]
- iii. If the distance between light source and metal P is doubled, how will the stopping potential change? [1]

OR

- a) A proton and electron have got same de- Broglie's wave length. Which has greater energy? [2]
- b) If the shortest wavelength in the Lyman series is 918°A, Calculate the longest wavelength in the same series. [3]

19. Millikan's oil drop experiment utilizes the electric field in a viscous fluid to analyze the speed of charged oil drop.

- What happens if a falling drop suddenly experience no net force in the vertical path? [1]
- What is the condition of upward terminal velocity of oil drop in Millikan's oil drop experiment? [1]

- Calculate the p.d. in volt necessary to be maintained between two horizontal conducting plates, one 5 mm above the other, so that a small oil drop of mass 1.31×10^{-14} kg with two electrons attached to it remains in equilibrium. ($g = 9.8 \text{ ms}^{-2}$, charge of electron = -1.6×10^{-19} C). [3]

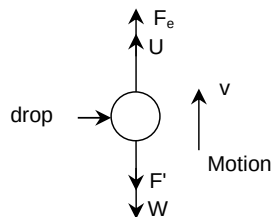


Fig- Upward motion of oil drop under electric field

Group - C

Give long answer to the following questions.

[3 × 8 = 24]

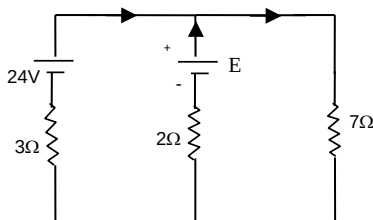
- Discuss Newton's formula for the velocity of sound in a gas medium with Laplace correction. [3]
 - A car, sounding a horn and producing a note of 500 Hz, approaches and then passes a stationary observer at a steady speed of 20m/s. Calculate the change in frequency heard by observer. (Velocity of sound=340m/s) [3]
 - Why does the empty vessel produce louder sound than filled one? [1]
 - Two organ pipes of same length open at both ends produce sound of different frequencies if their radii are different. Why? [1]

OR

- The time interval between flash of lightning and the sound of thunder is 2 seconds, when the temperature is 10°C . How far is the storm if the velocity of sound in air at 0°C is 330 m/s? [3]

- b) A whistle is revolving steadily in a horizontal circle. What changes in apparent pitch of the whistle will appear to a person standing outside the circle? Explain it by suitable diagram. If the person stands on the centre of circle, will there be change in pitch? Explain. [2+1]
- c) When we start filling an empty bucket with water, the pitch of sound produced goes on changing, why? [1]
- d) By what factor does the velocity of transverse wave in string change if its diameter is halved? [1]

21. a) State Ampere's circuital theorem and hence use it to find the magnetic field intensity due to a long current carrying solenoid. [1+3]
- b) When a charge particle moves in a uniform magnetic field, it experiences a force, called the Lorentz force. What is the vector representation of Lorentz force? [1]
- c) What must be the emf E in the current flowing through the 7Ω resistor is 1.80 A ? Each emf source has negligible internal resistance. [3]



OR

- a) A galvanometer is a device that measures or detects small currents with appropriate modification. It can be converted into an ammeter. Describe how can you convert a galvanometer into an ammeter. [3]
- b) Explain what do you mean by Seebeck Effect? How does thermoelectric e.m.f. varies with the temperature of hot junction? [3]
- c) The thermo-emf ' ε ' and the temperature of hot junction ' θ ' satisfy a relation $\varepsilon = a\theta + b\theta^2$ where $a = 4.1 \times 10^{-5}\text{V}(\text{°C})^{-1}$ and $b = -4.1 \times 10^{-8}\text{V}(\text{°C})^{-2}$. If the cold junction temperature is 0°C find the neutral temperature. [2]

22. a) Derive an expression for the total energy of an electron in n^{th} orbit of hydrogen atom, assuming Bohr's model to be valid. [3]
- b) An electron is in third excited state. How many photon wavelength are possible? [2]
- c) Calculate the wavelength of electromagnetic radiation emitted by a hydrogen atom which undergoes a transition between energy levels of $-1.36 \times 10^{-19} \text{ J}$ and $-5.45 \times 10^{-19} \text{ J}$.
(Given, Planck's constant = $6.6 \times 10^{-34} \text{ Js}$) [3]

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